

The Pythagorean Compendium: Complete Mathematical Specification

Philosophical Constructs, Computational Algorithms, and Harmonic Identities

Formal Data Export Services

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Abstract

This final edition of the Pythagorean Compendium provides the full suite of mathematical formulas identified in the data store. It bridges the gap between discrete integer geometry, continuous trigonometric identities, and recursive algorithmic generation.

1 Core Geometric Theorems

1.1 The Fundamental Theorem

The square of the hypotenuse (c) is equal to the sum of the squares of the legs (a, b):

$$a^2 + b^2 = c^2 \quad (1)$$

1.2 The Reciprocal Pythagorean Theorem

For a right triangle with legs a and b and altitude h dropped to the hypotenuse, the following relationship holds:

$$\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{h^2} \quad (2)$$

1.3 The Trigonometric Identity

Derived from the unit circle where the radius $r = 1$, the Pythagorean relationship is expressed in circular functions as:

$$\sin^2(\theta) + \cos^2(\theta) = 1 \quad (3)$$

2 Computational Generators

2.1 Euclid's Primitive Triple Generator

To ensure the generation of **Primitive** Pythagorean Triples, the following parameters are applied:

$$\begin{aligned} a &= m^2 - n^2 \\ b &= 2mn \\ c &= m^2 + n^2 \end{aligned}$$

Necessary and Sufficient Conditions:

- $m > n > 0$
- $\gcd(m, n) = 1$ (m and n are coprime)
- $m \not\equiv n \pmod{2}$ (one must be even, the other odd)

2.2 Side-Diagonal Recursion (Theon's Ladder)

The approximation of $\sqrt{2}$ is handled through a linear recurrence relation representing "Side" (s) and "Diagonal" (d) numbers:

$$\begin{cases} s_{n+1} = s_n + d_n \\ d_{n+1} = 2s_n + d_n \end{cases} \quad (4)$$

This can be expressed in matrix form as:

$$\begin{pmatrix} s_{n+1} \\ d_{n+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} s_n \\ d_n \end{pmatrix} \quad (5)$$

3 The Classical Means

The Pythagorean school identified the relationship between the Arithmetic (A), Geometric (G), and Harmonic (H) means. For two positive numbers a and b :

$$A = \frac{a+b}{2}, \quad G = \sqrt{ab}, \quad H = \frac{2ab}{a+b} \quad (6)$$

These means satisfy the proportion:

$$\frac{A}{G} = \frac{G}{H} \implies G^2 = AH \quad (7)$$

4 Ternary Tree of Primitive Triples

Every primitive triple can be generated by applying one of the three unimodular matrices M_1, M_2, M_3 to the seed triple $[3, 4, 5]^T$:

$$M_1 = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -1 & 2 \\ 2 & -2 & 3 \end{pmatrix}, M_2 = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{pmatrix}, M_3 = \begin{pmatrix} -1 & 2 & 2 \\ -2 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix} \quad (8)$$

5 Summary of Harmonic Ratios

Name	Ratio	Formula	Frequency Multiplier
Octave	2 : 1	2^1	2.000
Perfect Fifth	3 : 2	$3/2$	1.500
Perfect Fourth	4 : 3	$4/3$	1.333
Whole Tone	9 : 8	$(3/2)^2/2$	1.125
Pythagorean Comma	531441 : 524288	$(3/2)^{12}/2^7$	1.0136